

TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits

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ABSTRACT Localized plug-socket faults can progress to fire hazards through sustained overload, degraded-contact heating, and load-limited series arcing before conventional upstream protection responds. This paper presents a TinyML-enabled smart plug for complementary socket-level intervention in 230 V residential circuits. The prototype integrates custom current sensing, anti-aliasing and digitization, socket-temperature estimation from a contact-region thermistor, embedded Random Forest arc inference, deterministic overload and heating logic, event logging, and latched relay disconnection. Current frames were sampled at 27 kHz and converted to conduction-gap, baseline-departure, and spectral-disturbance features with load-context fields. Controlled validation gave mean absolute errors of 0.178 A for current and 1.02°C for temperature sensing. In overload tests, warning occurred in all heavy-load trials, while the sustained 17.30 A case tripped after 61.3 s. Degraded contacts heated faster than normal contacts under matched loads; the most severe protection trial tripped at an estimated 40.0°C while the reference terminal reached 56.0°C. Across 200 arc and normal trials, the deployed model-plus-firmware logic achieved 97.50% accuracy, 98.97% precision, 96.00% recall, and 97.46% F1-score. Results support prototype-scale feasibility for localized protective intervention, with broader field validation still required.

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1. INTRODUCTION

Electrical fires remain a persistent residential safety problem because multiple failure mechanisms can produce localized heating, arcing, or ignition before a simple breaker-level overcurrent condition is apparent (Meng et al. 2024; da Rocha et al. 2023; Taylor et al. 2024). At the outlet level, three fault classes are especially relevant to plug-connected appliances: sustained overload, poor-contact or degraded-contact heating, and load-limited series arcing. Poor plug-socket contact can raise local resistance and temperature, while thermally degraded insulation can support arc tracking at a plug-outlet connection (Zhang et al. 2021; Li et al. 2024; Hadžiefendić et al. 2018; Okamoto et al. 2022). These mechanisms are local, progressive, and strongly affected by contact condition, load current, materials, and thermal paths.

Conventional branch-circuit protection and certified arc-fault devices remain essential safety layers. However, upstream devices do not directly measure the thermal condition of a specific socket interface, and arc detection is difficult because normal appliances can produce nonlinear current, switching transients, motor noise, and high-frequency components that resemble fault signatures (Lee et al. 2000; Zhao et al. 2022; Gong et al. 2022; Chen et al. 2024). This motivates a complementary device located at the plug-socket interface, where current waveform behavior, local temperature, and relay intervention can be combined without cloud dependence.

This paper presents a TinyML Smart Plug for complementary socket-level protection against progressive fire-inducing faults in residential circuits. Its contribution is an integrated prototype that combines custom hardware, anti-aliasing and digitization, embedded feature extraction, local load-context and series-arc inference, deterministic overload and heating protection, event logging, and relay-based intervention. The device is evaluated as a controlled-laboratory research prototype, not as a certified replacement for breakers, AFCIs, GFCIs, electrical inspection, or code-compliant wiring.

2. MATERIALS AND METHODS

2.1 Prototype Hardware and Physical Design

Figure 1 summarizes the system architecture. The prototype was implemented as a plug-through 230 V single-phase smart plug. A custom hardware stack integrated the line-side plug, protected outlet, relay switching path, ESP32-S3 controller, current sensing, voltage context sensing, socket-temperature sensing, local indicators, enclosure, and logging interface. Voltage was retained only for waveform acquisition context, mains-state interpretation, and calibration; voltage-trip behavior is outside the scope of this article.

The current path used an ACS37035 Hall-effect sensor with its analog output routed through an active anti-aliasing stage and an MCP3204 external analog-to-digital



FIGURE 1. System architecture of the TinyML Smart Plug prototype. Protection decisions are executed locally on the ESP32-S3; cloud logging and the web interface support review, labeling, and monitoring.



FIGURE 2. Completed prototype enclosure showing the plug side and protected-outlet side used during bench validation. The enclosure documents the research form factor and is not a certification claim.

converter. The second-order low-pass stage was set near 10 kHz using

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \approx 9.98 \text{ kHz}, \quad (1)$$

for $R_1 = 200 \text{ k}\Omega$, $R_2 = 33 \text{ k}\Omega$, $C_1 = 470 \text{ pF}$, and $C_2 = 82 \text{ pF}$. The stage conditioned the current waveform before the firmware sampled 512-point frames for arc-feature extraction. A 10 k Ω NTC thermistor was placed near the plug-side contact region to provide a non-intrusive proxy for local socket heating. The relay was driven through a latched fault path so that a confirmed fault remained disconnected until reset.

Figure 2 shows the enclosed prototype. The physical design considered separation of mains and low-voltage wiring, airflow around internal components, access to the protected outlet, front-panel switching, buzzer and display placement, and repeatability during bench testing. The prototype is a laboratory platform; creepage, clearance, insulation barriers, relay rating, enclosure flame rating, and production manufacturability require separate compliance engineering before practical deployment.

2.2 Embedded Signal Processing and TinyML Workflow

The arc-detection path used 512-sample current frames at a 27 kHz target sampling rate with a 256-sample hop. A Hann window was applied before frequency-domain computations to reduce spectral leakage (Harris 1978). The firmware computed a broader set of waveform descriptors for logging and review, but the deployed series-arc Random Forest used six base waveform features: maximum low-current run, zero-dwell ratio, low-current ratio, absolute RMS-current z-score relative to baseline, midband residual ratio, and mid/high-frequency spectral flux. These features were selected to capture interruption, restrike, waveform departure, and spectral disturbance, which are recurring indicators in series-arc studies (Zhao et al. 2022; Gong et al. 2022; Chen et al. 2024; Qi et al. 2024).

Two representative deployed features are

$$L_{\text{ratio}} = \frac{100}{N} \sum_{n=0}^{N-1} \mathbf{1}(|i_c[n]| \leq I_{\text{low}}), \quad (2)$$

where $I_{\text{low}} = \max\{0.035, 0.06 \max(I_{\text{rms}}, 0.10)\}$, and

$$t_{\text{low,max}} = \frac{1000 N_{\text{run,max}}}{f_s}. \quad (3)$$

Here L_{ratio} expresses the percentage of near-zero-current samples, $i_c[n]$ is the cleaned current sample, $N_{\text{run,max}}$ is the longest low-current run, and f_s is the sampling rate. These equations directly represent the extinction-gap behavior expected in a series arc.

Load context was estimated separately from six current-waveform descriptors and encoded as six one-hot family fields plus a confidence value. The deployed arc model therefore used a 13-input vector: six waveform features and seven context fields. The embedded threshold was 0.4800 for supported known contexts, while unknown contexts used a more conservative threshold of 0.7800 and at least three corroborating feature votes. Random Forest classification was selected because tree ensembles can capture nonlinear feature interactions and can be exported as deterministic embedded code (Breiman 2001). This offline-training and on-device-inference workflow follows TinyML practice for constrained microcontrollers, where training remains off-device and inference runs locally at the edge (Ray 2021; Immonen and Hämäläinen 2022; Antonini et al. 2023).

The training workflow used firmware CSV logs from normal, startup, and induced-arc sessions. Arc windows were manually reviewed and labeled because true arc frames were sparse relative to normal load operation. Python scripts cleaned the logs, formed train/validation/test splits, trained the load-context and arc models, generated metric reports, and exported C headers for firmware integration. The final embedded behavior combined model output with a leaky persistence counter before relay tripping, so the reported device behavior represents model-plus-firmware protection rather than a standalone frame classifier.

The final arc data view contained 49,296 logged rows with 283 manually marked arc rows, or about 0.577% positives. The model-development split used 35,512 training rows, 9,434 validation rows, and 10,268 test rows. This imbalance motivated reporting both frame-level model results and trial-level relay outcomes.

2.3 Protection Logic and Test Procedures

Overload and socket-heating protection used deterministic logic because their dominant variables were current, time, and temperature. Series-arc protection used TinyML inference plus persistence logic to reduce nuisance trips from isolated frames. Table 1 summarizes the implemented functions.

The heating estimator used a squared current-shape

TABLE 1. Implemented hardware, sensing, and protection functions.

Block	Implementation	Role in evaluation
Current path	ACS37035 module, active anti-aliasing, MCP3204 ADC	RMS current, overload logic, waveform features
Thermal path	10 kΩ NTC near plug-side contact region	Socket-heating warning and relay trip estimate
Arc inference	13-input Random Forest exported to firmware	Series-arc classification with load context
Relay path	Latched relay intervention	Local disconnection after confirmed fault
Logging path	Firmware and web review tools	Dataset collection, manual labeling, validation records

term,

$$s_I = \left[\text{clamp} \left(\frac{I_{\text{rms}}}{12.0}, 0, 1.25 \right) \right]^2, \quad (4)$$

and selected the larger of a directly compensated thermistor value and an expected normal socket value:

$$T_{\text{socket,est}} = \text{clamp} \{ \max(T_{\text{NTC}} + \Delta T_{\text{meas}}, T_a + \Delta T_{\text{normal}}), -20, 125 \}. \quad (5)$$

In (4) and (5), T_a is the tracked baseline temperature, ΔT_{normal} is the filtered normal rise expected from load current, and $\Delta T_{\text{meas}} = 0.30 + 1.20s_I$ when a load is active. Socket-heating warning begins at 39.0°C, clears below 38.6°C, and trips at $T_{\text{socket,est}} \geq 40.0^\circ\text{C}$.

Overload persistence and arc persistence were expressed as leaky counters:

$$S_{\text{OL}}[k] = \begin{cases} \min(6, S_{\text{OL}}[k-1] + 1), & I_{\text{avg}} \geq 14.5 \text{ A}, \\ \max(0, S_{\text{OL}}[k-1] - 1), & I_{\text{avg}} \leq 14.1 \text{ A}, \end{cases} \quad (6)$$

updated every 10 s with relay trip at $S_{\text{OL}} = 6$, and

$$C_{\text{arc}}[k] = \begin{cases} \min(16, C_{\text{arc}}[k-1] + 2), & \hat{y}_{\text{arc}} = 1, \\ \max(0, C_{\text{arc}}[k-1] - 6), & \hat{y}_{\text{arc}} = 0, \end{cases} \quad (7)$$

with relay trip at $C_{\text{arc}} \geq 8$. The arc counter required repeated positive frames and decayed quickly when inference returned to normal.

Figure 3 shows the physical test setups retained for the journal article. Overload trials used high-power load combinations to verify warning and sustained-trip behavior. Socket-heating tests compared normal and intentionally degraded contacts under matched load cases, recording device temperature, estimated socket temperature, infrared terminal temperature, relay state, and elapsed time. Series-arc trials used a controlled series interruption platform with appliance loads; relay trip was the positive prototype response and an induced series arc was the positive ground-truth condition.

Classification performance was computed using accuracy, precision, recall, and F1-score from trial-level true-positive, false-positive, true-negative, and false-negative outcomes. The positive class was a relay trip during an arcing event. Sensor validation used mean absolute error against reference instruments.

3. RESULTS

Sensor validation supported the protection decisions used in the prototype. The calibrated current channel had a mean absolute error of 0.178 A, and the thermistor channel had a mean absolute error of 1.02°C at the sensor measurement point. The voltage channel produced 3.910 V mean absolute error and was retained for context and logging, but voltage-protection behavior is not reported here because the article focuses on fire-inducing socket faults.

The overload evaluation in Table 2 shows the intended separation between warning and relay disconnection. Trials with mean currents from 12.33 A to 14.42 A produced warnings but did not trip, even when peaks approached or exceeded 14.5 A. The relay opened only in Trial 4, where the mean current was 17.30 A and the 61.3 s latency agreed with the approximately 60 s persistence design.

Socket-heating validation showed that current magnitude alone was insufficient to describe outlet risk. Table 3 summarizes the protection trials under normal and corroded contacts. At approximately 9 A, the corroded socket reached the 38.2°C device-temperature marker in 48 min, compared with 100 min for the normal socket. At approximately 11 A, the corroded case reached the same marker in 7 min, compared with 54 min for the normal socket. In the most severe corroded case, the estimated socket temperature reached the 40.0°C trip threshold after 5 min while the infrared terminal reference reached 56.0°C.

Figure 4 presents representative thermal trends for the higher-current cases. The degraded contact heated faster and reached a larger terminal-temperature rise than the normal contact, while the external thermistor estimate lagged the hottest terminal region.

For series-arc detection, the complete deployed system detected 96 of 100 arc trials and produced one trip during 100 normal trials. Recognized loads accounted for 49 detected arcs, one missed arc, 50 correct normal outcomes, and no false trip. Unrecognized loads accounted for 47 detected arcs, three missed arcs, 49 correct normal outcomes, and one false trip. The residual errors therefore came mainly from more difficult or unrecognized load behavior, as shown in Table 4.

The exported Random Forest report provided a second view before full relay-level interpretation. In 10,268 held-out feature frames, the report contained 10,173 true negatives, 5 false positives, 25 false negatives, and 65 true positives. The corresponding frame-level accuracy was 99.7078%, precision was 92.8571%, recall was 72.2222%, and F1-score was 81.25%. These values were lower than the trial-level relay metrics because the report evaluated individual feature frames before persistence logic and trial-level interpretation.

TABLE 2. Overload protection results.

Trial	Load setup	Mean A	Peak A	Relay
1	Kettle + water heater	12.87	14.85	No
2	Two kettles	12.33	13.63	No
3	Kettle + water heater	13.36	14.86	No
4	Kettle + water heater + two lamps	17.30	19.73	Yes, 61.3 s
5	Kettle + water heater + one lamp	14.42	16.23	No



(a) Socket-heating validation

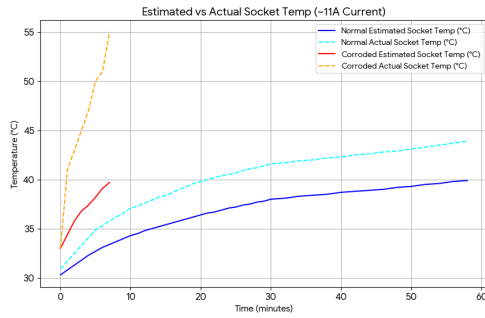


(b) Series-arc validation

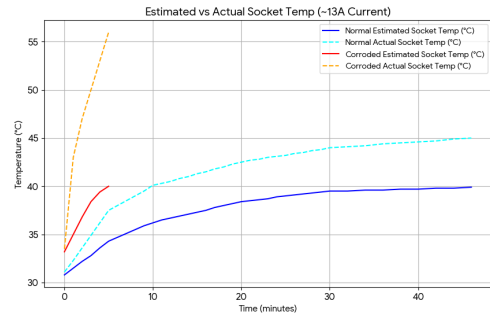
FIGURE 3. Representative bench setups for contact-heating and series-arc evaluation. The smart plug was placed in the load path so that current, temperature, relay state, and event logs were captured by the same embedded system used for protection.

TABLE 3. Socket-heating protection validation. Times are in minutes; dash indicates that the threshold was not reached before the end of the trial.

Load case	Socket	Mean A	Dev. 38.2°C	Est. 40°C	Est. temp.	IR temp.	Relay
Water heater	Normal	5.17	–	–	32.7	33.8	No
Water heater	Corroded	4.85	–	–	36.7	38.5	No
Water heater + 1 lamp	Normal	7.11	–	–	38.1	40.3	No
Water heater + 1 lamp	Corroded	7.10	100	–	39.8	42.4	Yes
Water heater + 2 lamps	Normal	9.06	100	120	40.0	43.2	Yes
Water heater + 2 lamps	Corroded	9.04	48	52	40.0	44.1	Yes
Water heater + 3 lamps	Normal	11.07	54	–	39.9	43.9	Yes
Water heater + 3 lamps	Corroded	11.14	7	–	39.7	55.0	Yes
Water heater + 4 lamps	Normal	13.02	44	–	39.9	45.0	Yes
Water heater + 4 lamps	Corroded	13.08	5	5	40.0	56.0	Yes



(a) Approximately 11 A



(b) Approximately 13 A

FIGURE 4. Representative socket-heating protection-validation trends under normal and corroded contacts.

4. DISCUSSION

The results support the feasibility of a local protective layer that acts closer to the plug-socket interface than panel-level protection. Overload behavior was intentionally conservative: the device warned for heavy load combinations and tripped only after sustained current stress accumulated. This behavior is suitable for a prototype intended to distinguish short transients from thermal overload stress.

The socket-heating result is the strongest justification for localized thermal supervision. Reduced contact force,

poor plug-receptacle contact, glowing contact, and arc tracking are established contributors to electrical fire risk (Hadžiefendić et al. 2018, 2019; Zhang et al. 2023; Okamoto et al. 2022). The prototype reproduced the practical concern in a controlled setting: degraded contacts heated faster and the hotter terminal region exceeded the device-side thermal estimate. The device-side thermistor is therefore useful as a protection-facing proxy, but it should not be interpreted as the true internal contact temperature.

The arc-detection results are encouraging but bounded. Current fluctuation, low-current behavior, zero-region features, and spectral disturbance are established indicators for series-arc detection (Zhao et al. 2022; Gong et al. 2022; Chen et al. 2024; Qi et al. 2024). The prototype converted these ideas into an embedded TinyML workflow by combining selected waveform features with load-family context. However, the remaining missed arcs and the single false trip occurred mainly in more difficult load behavior, which is consistent with load-monitoring literature

TABLE 4. Trial-level series-arc performance by load setup.

Load setup	Accuracy	Prec.	Recall	F1-score
Recognized	99.00%	100.00%	98.00%	98.99%
Unrecognized	96.00%	97.92%	94.00%	95.92%
All tested loads	97.50%	98.97%	96.00%	97.46%

showing that appliance current signatures vary across load families and models (Medico et al. 2020; Shareef et al. 2023).

Several limitations remain. The experiments were controlled and prototype-scale, not long-term household field trials. The load and fault set did not cover all appliances, contact materials, electrode gaps, wiring conditions, nuisance transients, or ambient conditions. The thermal estimator used a non-intrusive sensor location and can under-read internal hotspots. The model also depends on dataset diversity; expanding normal, transient, nuisance, and arc examples is necessary before claims of consumer-ready reliability or standards equivalence can be made. The device should therefore be viewed as a complementary protective platform rather than an AFCI-equivalent product.

5. CONCLUSIONS

This study demonstrated the controlled-laboratory feasibility of a TinyML-enabled smart plug for local intervention against sustained overload, series-arc, and degraded-contact heating risks. The prototype combined custom sensing hardware, current-waveform acquisition, socket-temperature estimation, feature engineering, load-context inference, Random Forest arc classification, deterministic overload and heating protection, and relay-based local disconnection. The strongest results were sustained-overload tripping after 61.3 s in the highest-current case, faster and hotter degraded-socket heating under comparable load, and 97.50% trial-level series-arc accuracy across 200 arc and normal outcomes.

Future work should expand the arc and nuisance-load datasets, validate the thermal estimator across additional sockets and contact materials, improve sensor placement or contact-temperature modeling, and conduct longer-term household-style trials. The prototype is best interpreted as a complementary local protection layer, not a device that fully prevents all electrical fires.

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AUTHORS' CONTRIBUTIONS

FMP led the design, implementation, data analysis, TinyML workflow, validation, and manuscript preparation. JML designed the ACS37035 current-sensor module, helped initialize the first test bench, assisted component procurement, contributed to data collection, and provided formatting and technical inputs. AAA refined the test bench, contributed to data collection, and provided manuscript inputs. RDS supervised the research direction, methodology, and manuscript review. All authors read and approved the final manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

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